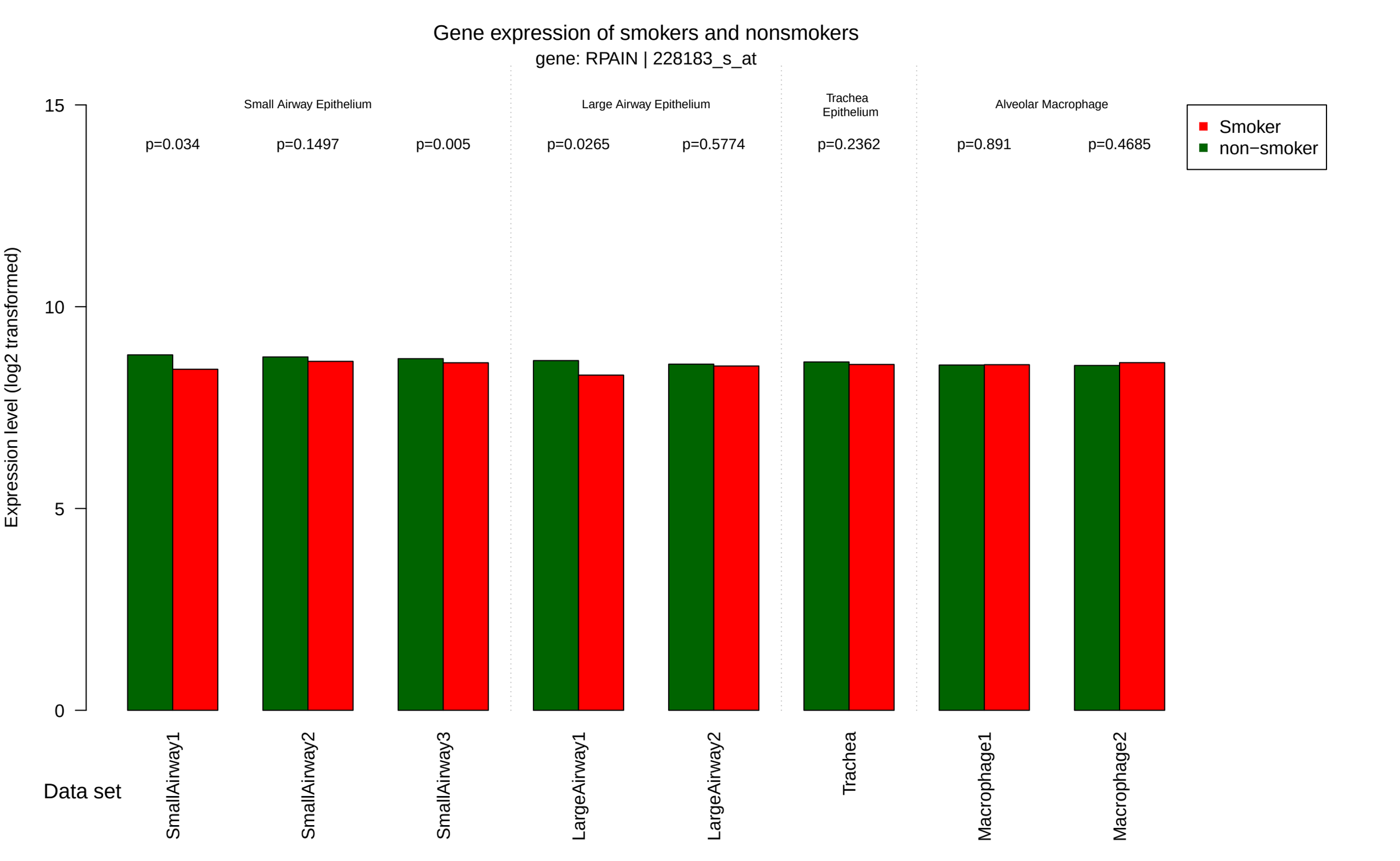
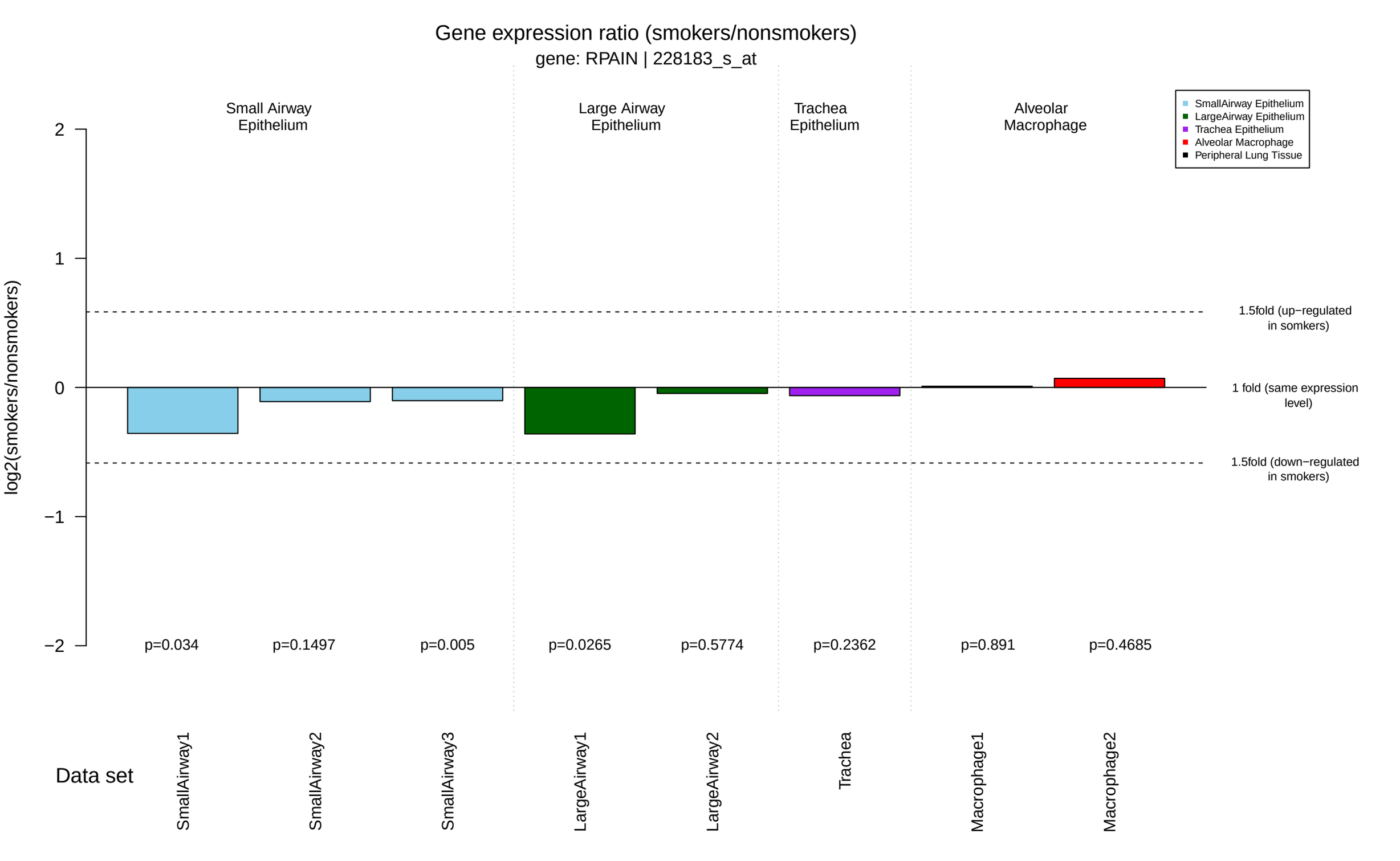
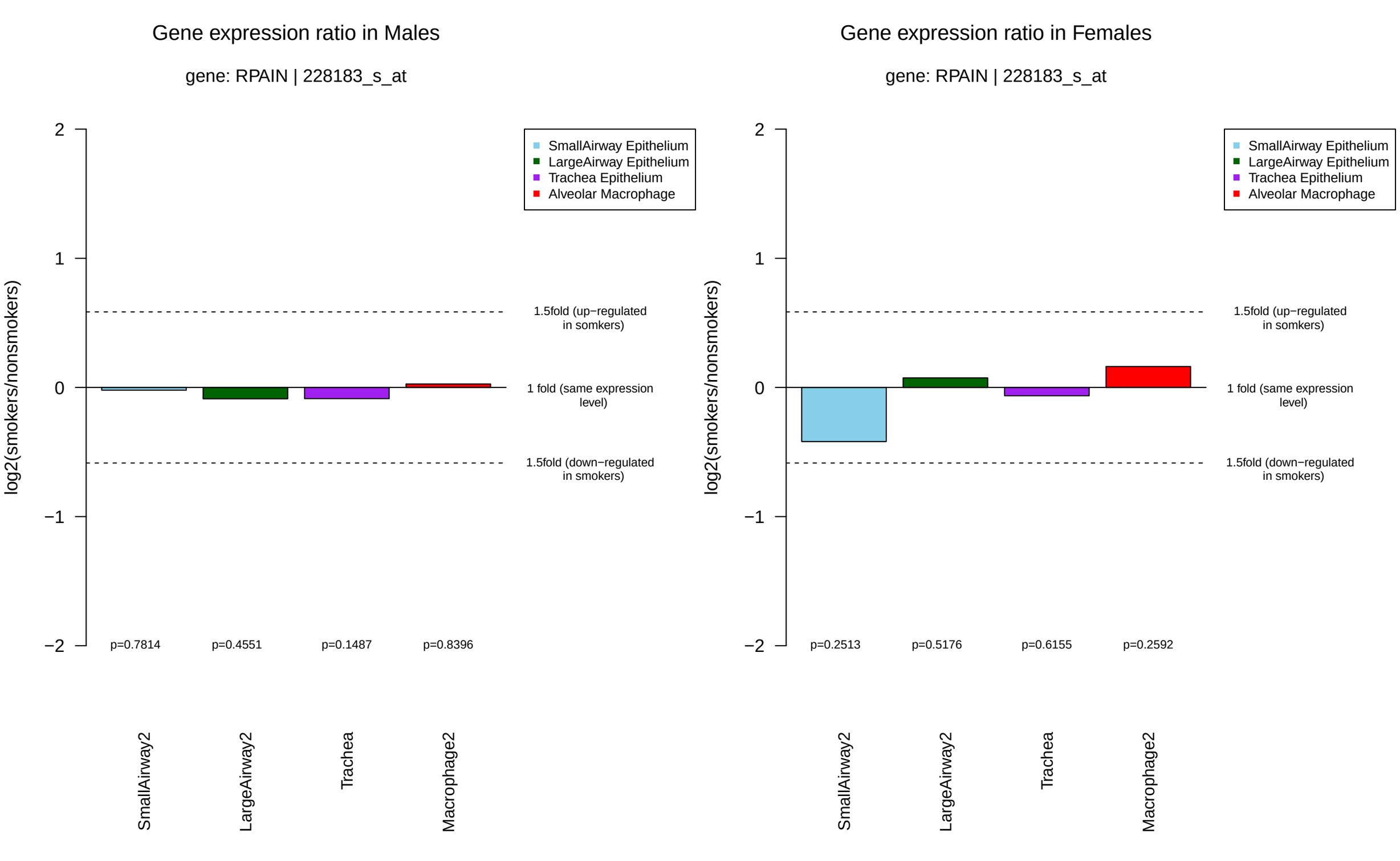
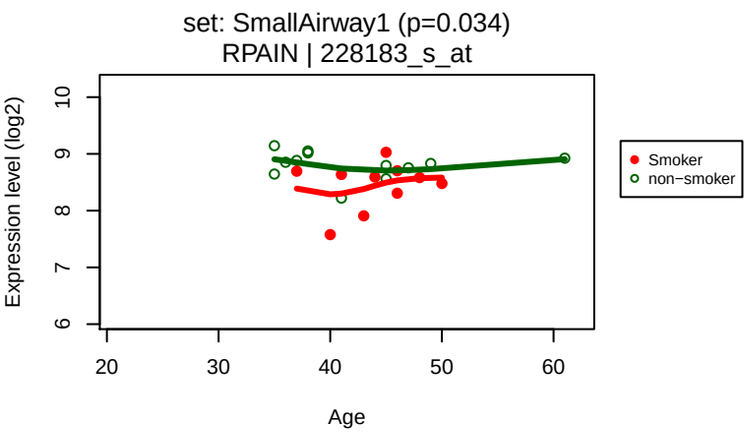
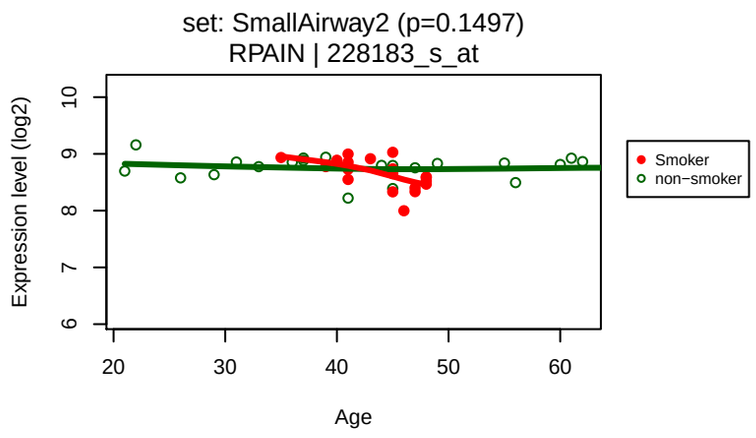
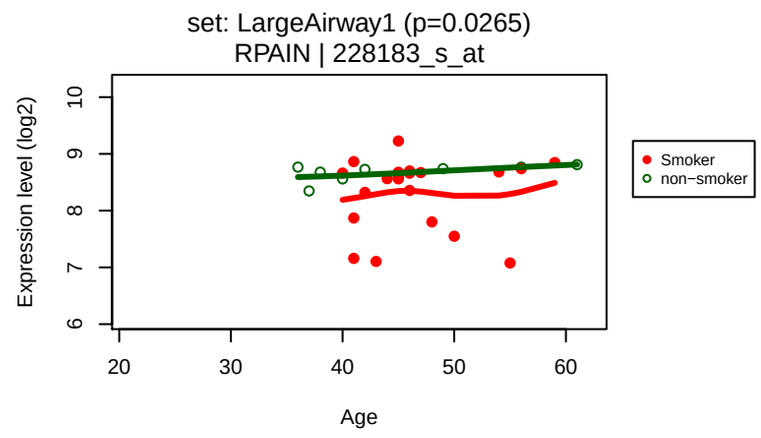

Gene expression of smokers and nonsmokers
gene: RPAIN | 228183_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.034
p=0.1497
p=0.005
p=0.0265
p=0.5774
p=0.2362
p=0.891
p=0.4685
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

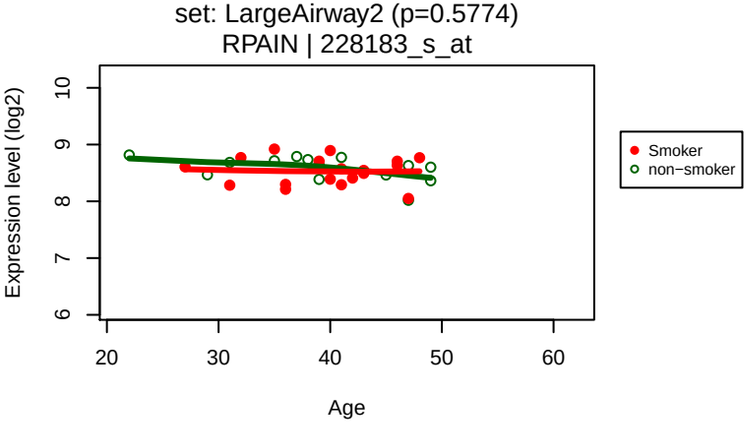

Gene expression ratio (smokers/nonsmokers)
gene: RPAIN | 228183_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.034
p=0.1497
p=0.005
p=0.0265
p=0.5774
p=0.2362
p=0.891
p=0.4685
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

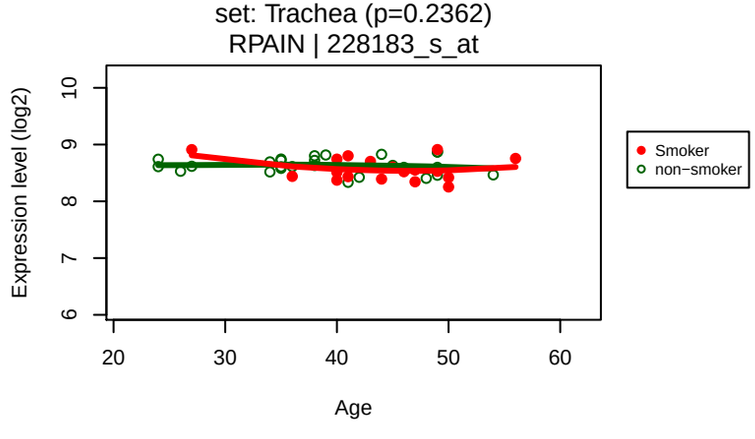

Gene expression ratio in Males
gene: RPAIN | 228183_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7814
p=0.4551
p=0.1487
p=0.8396
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPAIN | 228183_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2513
p=0.5176
p=0.6155
p=0.2592
SmallAirway2
LargeAirway2
Trachea
Macrophage2

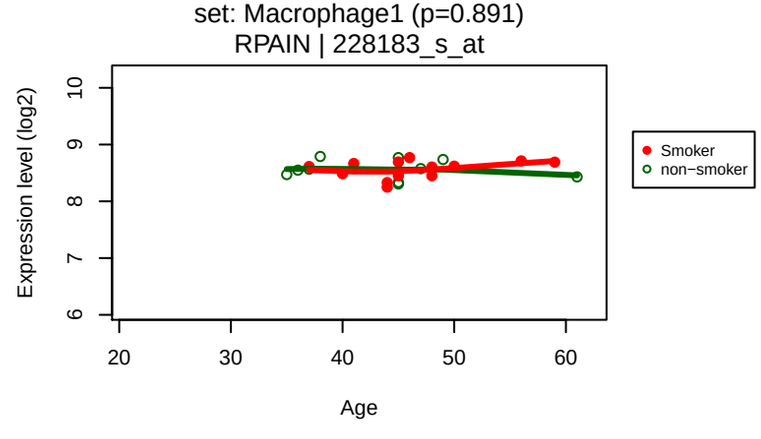

set: SmallAirway1 (p=0.034)
RPAIN | 228183_s_at
Expression level (log2)
Age
Smoker
non−smoker

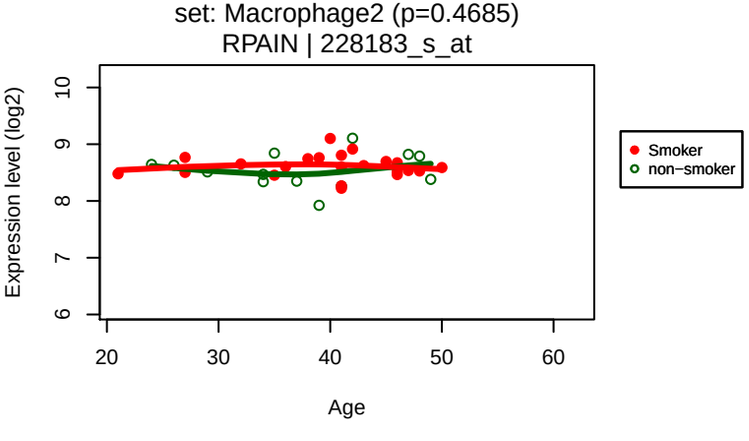

set: SmallAirway2 (p=0.1497)
RPAIN | 228183_s_at
Expression level (log2)
Age
Smoker
non−smoker

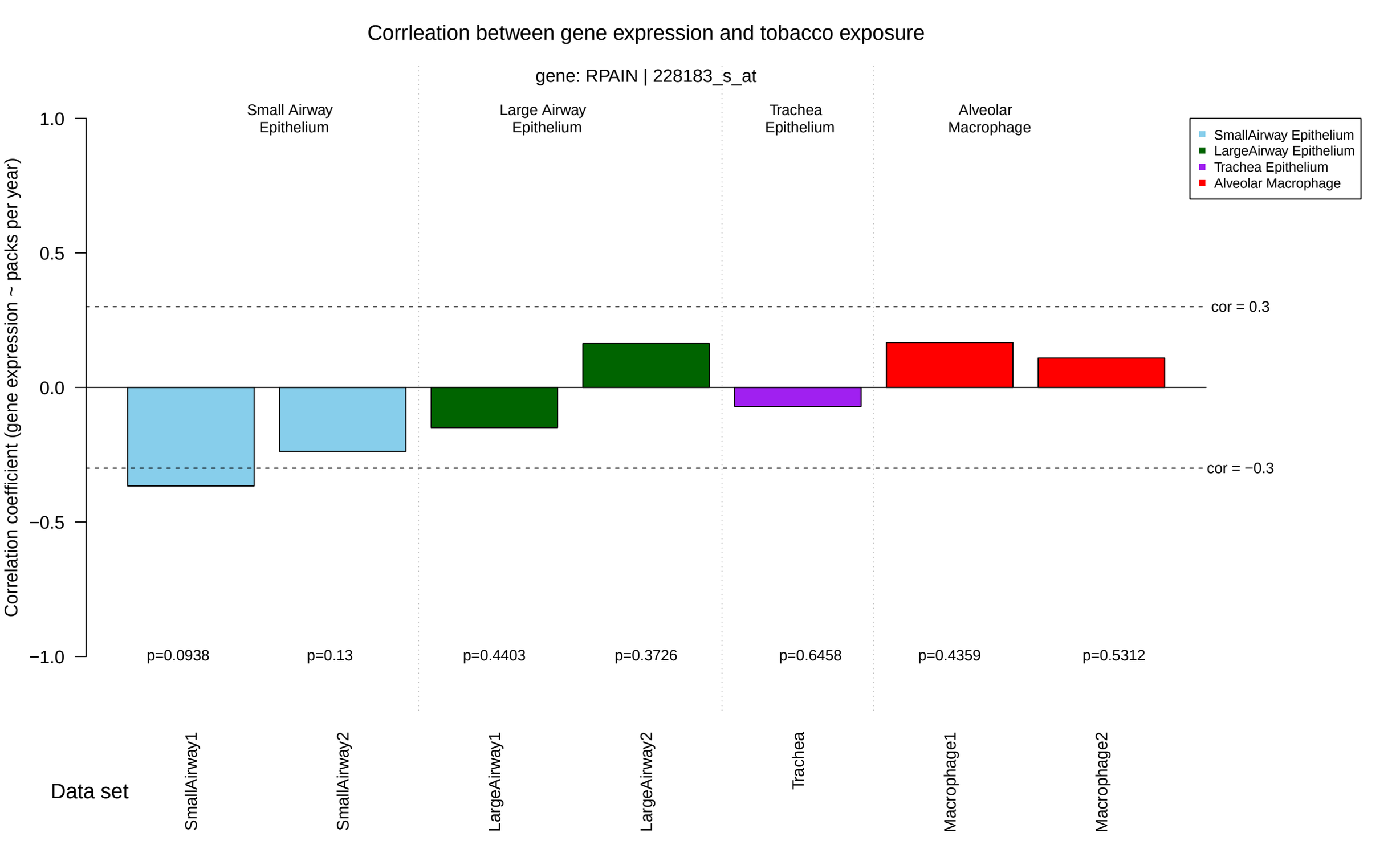

set: LargeAirway1 (p=0.0265)
RPAIN | 228183_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5774)
RPAIN | 228183_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2362)
RPAIN | 228183_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.891)
RPAIN | 228183_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4685)
RPAIN | 228183_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPAIN | 228183_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0938
p=0.13
p=0.4403
p=0.3726
p=0.6458
p=0.4359
p=0.5312
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2